

NIFTY100 Financial Intelligence Platform

REST API Documentation

Framework: FastAPI

Response Format: JSON

Database: SQLite

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Purpose

This document provides comprehensive documentation for the REST API of the NIFTY100 Financial Intelligence Platform. It describes available endpoints, request methods, parameters, response formats, error handling, and testing procedures.

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Introduction

The REST API exposes financial data from the NIFTY100 Financial Intelligence Platform, enabling external applications, dashboards, and analytical tools to access company information programmatically.

The API follows REST principles and returns data in JSON format.

Features

The API supports:

- Company information retrieval
- Financial statement access
- Stock screening
- Sector analysis
- Valuation metrics
- Portfolio analytics
- Report generation
- Health monitoring

Technology Stack

Component	Technology
Framework	FastAPI
Language	Python
Database	SQLite
Response Format	JSON
Documentation	Swagger UI / ReDoc

Base URL

Local Development

<http://localhost:8000>

Swagger UI

<http://localhost:8000/docs>

ReDoc

<http://localhost:8000/redoc>

HTTP Methods

Method	Purpose
GET	Retrieve data
POST	Submit filters or parameters
PUT	Update records (if supported)
DELETE	Remove records (if supported)

Running the API

```
uvicorn src.api.main:app --reload
```

Expected Output

```
INFO: Uvicorn running on http://127.0.0.1:8000
```

API Workflow

Client Request

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FastAPI Router

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Business Logic

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SQLite Database

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JSON Response

Response Structure

Successful responses:

```
{  
  "success": true,  
  "data": {  
    ...  
  }  
}
```

Failed responses:

```
{  
  "success": false,  
  "message": "Company not found"  
}
```

Content Type

All requests and responses use:

Content-Type: application/json

API Version

Current Version

v1

System Endpoints

System endpoints provide information about the health and availability of the API service. These endpoints are typically used by monitoring systems, deployment pipelines, and developers to verify that the service is running correctly.

Health Check

Endpoint

GET /health

Description

Returns the current health status of the API server.

Request

```
curl http://localhost:8000/health
```

Successful Response

```
{  
  "status": "healthy",  
  "database": "connected",  
  "version": "1.0.0"  
}
```

Status Codes

Code Description

200 API is healthy

500 Internal server error

API Information

Endpoint

GET /

Description

Returns basic information about the API including version, available modules, and documentation links.

Request

```
curl http://localhost:8000/
```

Example Response

```
{  
  "application": "NIFTY100 Financial Intelligence API",  
  "version": "1.0",  
  "documentation": "/docs",  
  "status": "running"  
}
```

Page 5 — Company Endpoints

3. Company APIs

These endpoints provide detailed information about individual companies stored in the database.

Get All Companies

Endpoint

GET /companies

Description

Returns a list of all companies available in the platform.

Request

```
curl http://localhost:8000/companies
```

Example Response

```
[  
  {
```

```
[
  {
    "ticker": "TCS",
    "company_name": "Tata Consultancy Services",
    "sector": "Information Technology"
  },
  {
    "ticker": "INFY",
    "company_name": "Infosys Limited",
    "sector": "Information Technology"
  }
]
```

Get Company Details

Endpoint

GET /company/{ticker}

URL Parameter

Parameter	Type	Description
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ticker	String	Stock ticker symbol
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Example Request

```
curl http://localhost:8000/company/TCS
```

Example Response

```
{
  "ticker": "TCS",
  "company_name": "Tata Consultancy Services",
  "sector": "Information Technology",
  "market_cap": 1580000000000,
  "roe": 42.1,
  "roce": 58.7,
```



```
"pe_ratio": 29.6
}
```

Status Codes

Code Meaning

200 Success

404 Company not found

500 Internal server error

Search Companies

Endpoint

GET /search

Query Parameters

Parameter Required Description

q	Yes	Company name or ticker
---	-----	------------------------

Example

```
curl "http://localhost:8000/search?q=TCS"
```

Page 6 — Financial Screener Endpoints

4. Financial Screener API

The screener endpoint allows users to filter companies based on financial criteria.

Run Stock Screener

Endpoint

POST /screener

Description

Returns companies matching specified screening conditions.

Request Body

```
{  
  "roe": 20,  
  "roce": 25,  
  "pe_ratio": 30,  
  "debt_equity": 1  
}
```

cURL Example

```
curl -X POST http://localhost:8000/screener \  
-H "Content-Type: application/json" \  
-d '{  
  "roe":20,  
  "roce":25,  
  "pe_ratio":30,  
  "debt_equity":1  
'
```

Example Response

```
{  
  "total_results": 8,  
  "companies": [  
    {  
      "ticker": "TCS",  
      "roe": 42.1,  
      "pe_ratio": 29.6  
    },  
    {  
      "ticker": "INFY",
```

```
      "roe": 30.5,  
      "pe_ratio": 27.1  
    }  
  ]  
}
```

Validation Rules

- $\text{ROE} \geq 0$
- $\text{ROCE} \geq 0$
- $\text{Debt-to-Equity} \geq 0$
- $\text{P/E Ratio} \geq 0$

Status Codes

Code Description

200 Success

400 Invalid filter values

500 Internal server error

Sector APIs

Sector APIs provide aggregated information for companies grouped by industry.

Get Sector Summary

Endpoint

GET /sector/{sector_name}

Example Request

```
curl http://localhost:8000/sector/Information%20Technology
```

Example Response

```
{
  "sector": "Information Technology",
  "companies": 8,
  "average_roe": 27.4,
  "average_pe": 30.2,
  "market_cap": 3950000000000
}
```

List All Sectors

Endpoint

GET /sectors

Example

```
curl http://localhost:8000/sectors
```

Example Response

```
[
  "Banking",
  "Information Technology",
  "Automobile",
```

```
"Pharmaceuticals",  
"Energy",  
"FMCG"  
]
```

Company Comparison

Endpoint

POST /compare

Example Request

```
{  
  "companies": [  
    "TCS",  
    "INFY",  
    "WIPRO"  
  ]  
}
```

Example Response

```
{  
  "comparison": [  
    {  
      "ticker": "TCS",  
      "roe": 42.1  
    },  
    {  
      "ticker": "INFY",  
      "roe": 30.4  
    },  
    {
```

```
    "ticker": "WIPRO",  
    "roe": 19.6  
  }  
]  
}
```

Valuation APIs

These endpoints return company valuation metrics and portfolio analytics.

Get Company Valuation

Endpoint

GET /valuation/{ticker}

Example Request

```
curl http://localhost:8000/valuation/TCS
```

Example Response

```
{
  "ticker": "TCS",
  "pe_ratio": 29.6,
  "pb_ratio": 12.5,
  "enterprise_value": 1645000000000,
  "ev_ebitda": 19.4,
  "recommendation": "Buy"
}
```

Portfolio Summary

Endpoint

GET /portfolio

Description

Returns summary statistics for the configured portfolio.

Example Response

```
{
  "total_companies": 10,
```

```
"expected_return": 15.2,  
"portfolio_risk": 9.8,  
"diversification_score": 87  
}
```

Portfolio Optimization

Endpoint

POST /portfolio/optimize

Request

```
{  
  "budget": 100000,  
  "risk_level": "Moderate"  
}
```

Example Response

```
{  
  "recommended_allocation": [  
    {  
      "ticker": "TCS",  
      "weight": 25  
    },  
    {  
      "ticker": "HDFCBANK",  
      "weight": 20  
    }  
  ]  
}
```


Endpoint Summary

Endpoint	Method	Description
/health	GET	Health check
/companies	GET	List companies
/company/{ticker}	GET	Company details
/search	GET	Search companies
/screener	POST	Financial screener
/sector/{sector}	GET	Sector summary
/sectors	GET	List sectors
/compare	POST	Compare companies
/valuation/{ticker}	GET	Company valuation
/portfolio	GET	Portfolio summary
/portfolio/optimize	POST	Portfolio optimization